11.9 A Compact 12V-to-1V 91.8% Peak Efficiency Hybrid Resonant Switched-Capacitor Parallel Inductor (ReSC-PL) Buck Converter

Guigang Cai[1], Yan Lu[1], Rui P. Martins[1,2]

[1]University of Macau, Macau, China
[2]Instituto Superior Tecnico/University of Lisboa, Lisbon, Portugal

High power density, high efficiency, and high voltage conversion ratio (VCR) DC-DC converters are in great demand for portable devices and autonomous robots. The double-step-down (DSD) converter and its variants are promising solutions for their reduced voltage stress on the power transistors, extended duty-cycle (D), and higher equivalent switching frequency [1–4]. However, all these topologies suffer from the trade-off between efficiency and current density, since all the output current I_O flows through the power inductors, we have to either use bulky inductors for high efficiency or use small inductors for high current density. To deal with this, [5] proposed an always-dual-path (ADP) hybrid converter that can reduce the inductor current I_L to $0.5I_O$, but at the cost of a larger inductor current ripple ΔI_L. Besides, [6] proposed a hybrid switched-capacitor (SC)-parallel-inductor (CPL) buck converter that reduces I_L without enlarging ΔI_L, achieving high efficiency and high current density simultaneously. Nevertheless, such CPL-Buck converter only deals with a 1-cell battery input, and is not suitable for high VCR conversion, e.g., 12V-to-1V. Because its inductor current reduction effect highly depends on D, and thus its benefits drop heavily at a high VCR. In this work, we propose a hybrid resonant SC parallel inductor (ReSC-PL) buck converter that can always effectively reduce I_L to below $0.5I_O \sim 0.67I_O$ for high VCRs ranging from 10 to 20.

Figure 11.9.1 shows the topology of the proposed resonant switched-capacitor parallel inductor (ReSC-PL) buck converter. It mainly consists of 12 switches M_{1-12}, 4 flying capacitors C_{F1-4}, one main inductor L and one parasitic inductor L_P. The proposed ReSC-PL buck converter can be divided into two parts: one high side SC to lower down the voltage of V_{X5}; one low side resonant SC to offer a second current path to reduce I_L. The idea of high side SC arises from a 4-level flying capacitor multilevel (FCML) converter which can constantly reduce the switching node voltage V_X to $V_{IN}/4$, while the CPL buck converter only reduces its V_X to $V_{IN}-2V_{OUT}$ which is VCR-related. The high side SC consists of 7 switches M_{1-7} and 2 flying capacitors C_{F1} and C_{F2}. It can operate in three modes: 1/4×, 1/3×and 1/2× modes, generating a V_{X5} of $V_{IN}/4$, $V_{IN}/3$ and $V_{IN}/2$, respectively. A few nH L_P in the low side resonant SC can not only reduce the output glitches coupled by high dV/dt in V_{X5} but also can make C_{F3} and C_{F4} operate in resonance at a proper frequency.

Figure 11.9.2 exhibits the operating principle of the ReSC-PL buck converter. One advantage is that the high side and low side operate separately. There are two states for the high side SC: ON-state (0 to DT) and OFF-state (DT to T). In the ON-state, M_8 is off and C_{F3} is charged in series with L through V_{X5}, where V_{X5} is equal to $V_{IN}/4$, $V_{IN}/3$ or $V_{IN}/2$ according to the operating mode. In the 1/4× mode, V_{X5} is equal to $V_{IN}/4$. There are four phases Φ_{H1-4} with $V_{CF1}=V_{IN}/2$ and $V_{CF2}=V_{IN}/4$. In Φ_{H1}, V_{IN} charges C_{F1} and C_{F2} in series; in Φ_{H2}, C_{F2} discharges to V_{X5}; in Φ_{H3}, C_{F1} charges C_{F2}; Φ_{H4} is the same as Φ_{H2}. In the 1/3× mode, V_{X5} is equal to $V_{IN}/3$. There are three phases Φ_{H1-3} with $V_{CF1}=2V_{IN}/3$ and $V_{CF2}=V_{IN}/3$. In Φ_{H1}, V_{IN} charges C_{F1}; in Φ_{H2}, C_{F1} charges C_{F2}; in Φ_{H3}, C_{F2} discharges to V_{X5}. In the 1/2× mode, V_{X5} is equal to $V_{IN}/2$. There are two phases Φ_{H1-2} with $V_{CF1}=V_{IN}/2$, where C_{F2} is floating. In Φ_{H1}, V_{IN} charges C_{F1}; in Φ_{H2}, C_{F1} discharges to V_{X5}. In the OFF-state, M_8 is on and L gets de-energized, and C_{F1} and C_{F2} are floating.

For the low side resonant SC, there are three phases, Φ_{L1} (0 to 0.5T), Φ_{L0} (0.5 to 0.6T) and Φ_{L2} (0.6 to T). In Φ_{L1}, C_{F4} discharges in series with L_P; in Φ_{L0}, C_{F4} is floating; in Φ_{L2}, C_{F3} charges C_{F4}. Notice that the operation of the low side resonant SC is independent of D. The durations of Φ_{L1} and Φ_{L2} are designed to 0.4T and 0.5T, respectively, to match the LC resonant periods nesting C_{F3}, C_{F4} and L_P, that is why we have a resting phase Φ_{L0} here. Then, the resonant frequencies decide the switching frequency of this work.

The bottom left of Fig. 11.9.2 shows the timing diagram of the 1/4× mode. Due to the high side SC and the low side resonant SC, the average inductor current I_{L_AVG} effectively reduces to less than $0.67I_O$ for a 12V input and 0.6-to-1.2V output, while the I_{L_AVG} reduction is only 10%-20% for the CPL buck, as shown in Fig. 2(bottom right). This indicates a great reduction in the inductor conduction loss. In addition, lowering the V_X (V_{x5}) voltage also means a smaller switching loss.

Figure 11.9.3 displays the schematic of the low side resonant SC with parasitic inductors. Here, the parasitic inductors are estimated according to the bond wire, capacitors' ESL and PCB trace of this prototype. High dV/dt of V_{X5} will induce glitches on V_{OUT} through the drain-source parasitic capacitor of M_9, when M_{11} is on. Therefore, in this prototype, we intentionally enlarge the parasitic inductor (L_{PO}) between the on-chip output and the PCB output V_{OUT}. Simulation results show that a few nH L_{PO} is large enough to reduce the glitches on V_{OUT} as given in Fig. 11.9.3. In addition, this L_{PO}, together with other parasitic inductors from C_{F3} and C_{F4} form a resonant SC operation. Since the equivalent inductance is a quite small number, when compared with the equivalent resistance that mainly comes from the switches' on-resistor and the bond wire resistor, the low side resonant operation is a relatively low-quality factor (low-Q) resonant. As shown in Fig. 11.9.3 (bottom right), when compared with a high-Q resonant current, the low-Q resonant currents I_{SC1} and I_{SC2} have higher peak values for the same average current. However, it has one benefit: the turn-off current is not so sensitive to the switching frequency. If M_9 and M_{11} turn off earlier at t_1, the residual current of I_{SC2} is smaller than that of the high-Q resonant current. This feature lowers down the demand for an accurate zero current detect circuit. Besides, though a relatively large parasitic inductance will help to form the resonant operation, the parasitic inductances of C_{F3}, as well as C_{F1} and C_{F2}, should be as small as possible because these capacitors are charged with I_L, not the resonance current. A large parasitic inductance will induce sever ringing on the switching nodes. On the other hand, we can design the parasitic inductance of C_{F4}, especially L_{PO}, relatively larger, because they only deal with resonance currents.

This design, fabricated in a 0.18μm BCD process, occupies an area of 8.88mm^2. Figure 11.9.4 plots the measured steady-state waveforms and load transient response. The proposed ReSC-PL buck converter operates in the 1/4×, 1/3×, and 1/2× modes for I_O=1A, and V_{OUT}=0.6V, 1V, and 1.2V, respectively. Meanwhile, I_{L_AVG} are reduced to 550mA, 450mA, and 550mA, respectively, showing I_L reductions of 45%, 55%, and 45%. In addition, during the high side SC ON-state, we can lower down V_{X5} to 3V, 4V and 6V, respectively. The measured load transient verifies the stability of this design. The V_{OUT} ripples and glitches are close to 65mV and 45mV, respectively, for 5A I_O, with 100μF C_O. We measured the efficiency under 0.6-to-1.2V V_{OUT}. From Fig. 11.9.5, the proposed ReSC-PL buck converter achieves the peak efficiency of 91.8% with V_{OUT}=1V and I_O=1A, and 87.9% with V_{OUT}=0.6V and I_O=1A. Resulting from the I_L reduction, we can use a 1μH inductor with a size as small as 2.5×2×1.2mm^3 to handle a maximum I_O of 5A while still holding high efficiency. When compared with prior arts [1–4], this work achieves the highest peak efficiency and the second highest current density as shown in Fig. 11.9.5 (bottom left). Furthermore, as compared to [1–4], this work achieves the highest peak efficiency for a VCR range of 10 to 20, as shown in Fig. 11.9.5 (bottom right).

Figure 11.9.6 presents the performance summary and comparison with state-of-the-art designs. We chose a switching frequency of 0.8MHz based on the 10μF C_{F3} and C_{F4} and the estimated L_P. This work, supporting a 5A output current (second only to [1]), uses only one (also the smallest) inductor, and achieves the highest peak efficiency. Though this work uses more capacitors, the total passive volume, including the inductors and flying capacitors, is still the smallest. Therefore, it achieves a current density of 350A/cm^3, second only to [2], but with 6.8% higher peak efficiency. Figure 11.97 shows the chip micrograph with C_{F3} attached on-die, and the PCB solution. In conclusion, by cascading a high side SC to reduce the switching node voltage and a low side resonant SC to further reduce the switching node voltage and reduce the inductor current, the proposed ReSC-PL buck converter reduces the inductor current for a high and wide VCR range, achieving high efficiency and high current density.

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References:

[1] Texas Instruments, "Data Sheet: TPS54a20, 8-V to 14-V Input, 10-A, up to 10-MHz SWIFT™ Step Down Converter," April 2016. Accessed: Dec. 1, 2022. <http://www.ti.com/lit/ds/symlink/tps54a20.pdf>.
[2] K. Wei et al., "A Direct 12V/24V-to-1V 3W 91.2%-Efficiency Tri-State DSD Power Converter with Online VCF Rebalancing and In-Situ Precharge Rate Regulation," *ISSCC*, pp. 190–191, Feb. 2020.
[3] J. Yuan et al., "A 12V/24V-to-1V DSD Power Converter with 56mV Droop and 0.9μs 1% Settling Time for a 3A/20ns Load Transient," *ISSCC*, pp. 300–301, Feb. 2022.
[4] T. Hu et al., "A 4A 12-to-1 Flying Capacitor Cross-Connected DC-DC Converter with Inserted D>0.5 Control Achieving> 2× Transient Inductor Current Slew Rate and 0.73× Theoretical Minimum Output Undershoot of DSD," *ISSCC*, pp. 302–303, Feb. 2022.
[5] J. Ko et al., "A 4.5 V-Input 0.3-to-1.7 V-Output Step-Down Always-Dual-Path DC-DC Converter Achieving 91.5%-Efficiency with 250mΩ-DCR Inductor for Low-Voltage SoCs," *IEEE Symp. VLSI Circuits*, pp. 1–2, Jun. 2021.
[6] G. Cai et al., "A Battery-Input Sub-1V Output 92.9% Peak Efficiency 0.3 A/mm² Current Density Hybrid SC-Parallel-Inductor Buck Converter with Reduced Inductor Current in 65nm CMOS," *ISSCC*, pp. 312–313, Feb. 2022.

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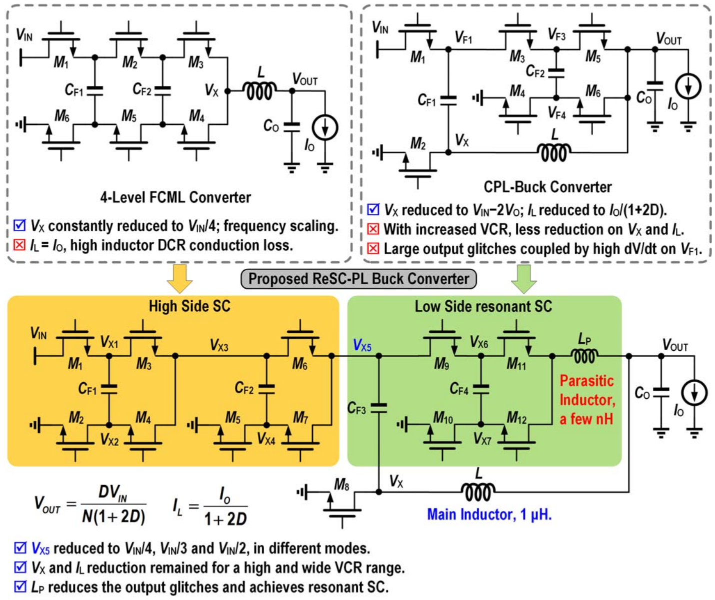


Figure 11.9.1: Topology of the proposed ReSC-PL buck converter.

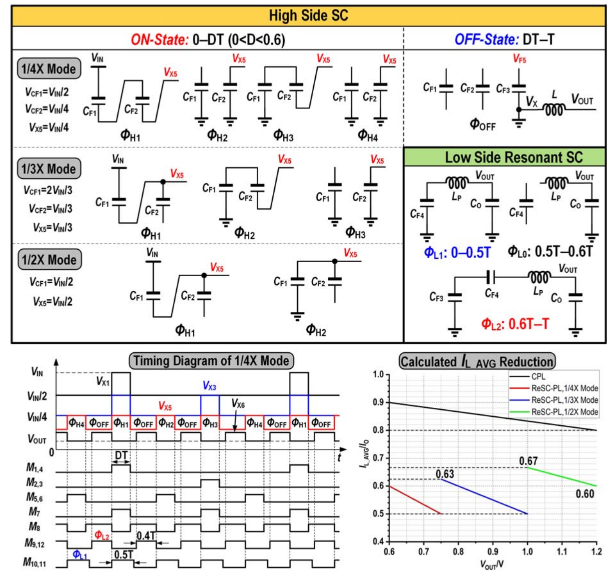


Figure 11.9.2: Operating principle of the proposed ReSC-PL buck converter, timing diagram of the 1/4× mode, and calculated I_{L_AVG} reduction.

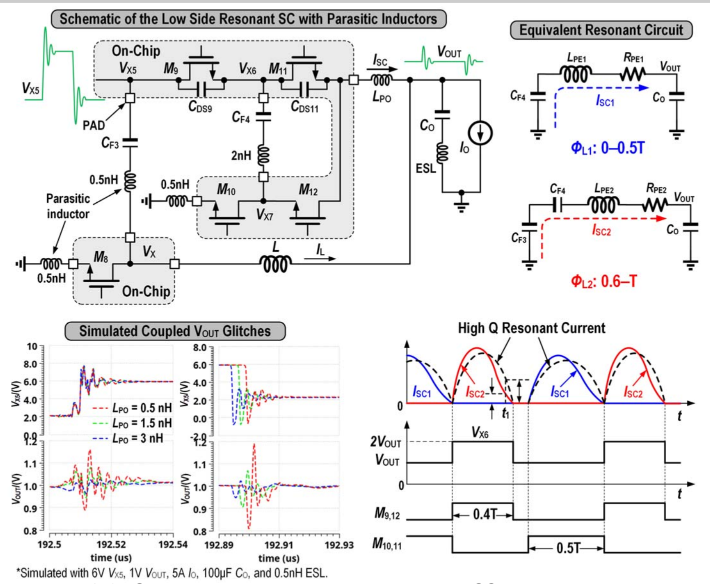


Figure 11.9.3: Schematic of the low side resonant SC with parasitic inductor, equivalent circuits, and timing diagram.

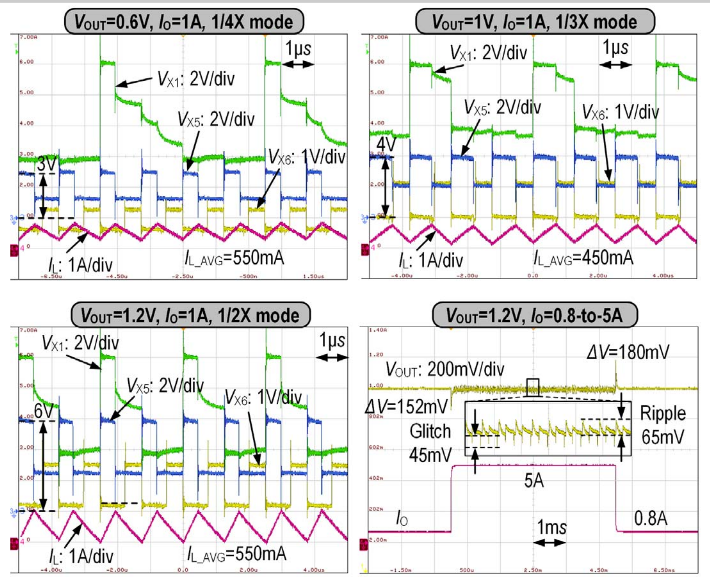


Figure 11.9.4: Measured steady state waveforms of the 1/4× mode, the 1/3× mode, the 1/2× mode and load transient response.

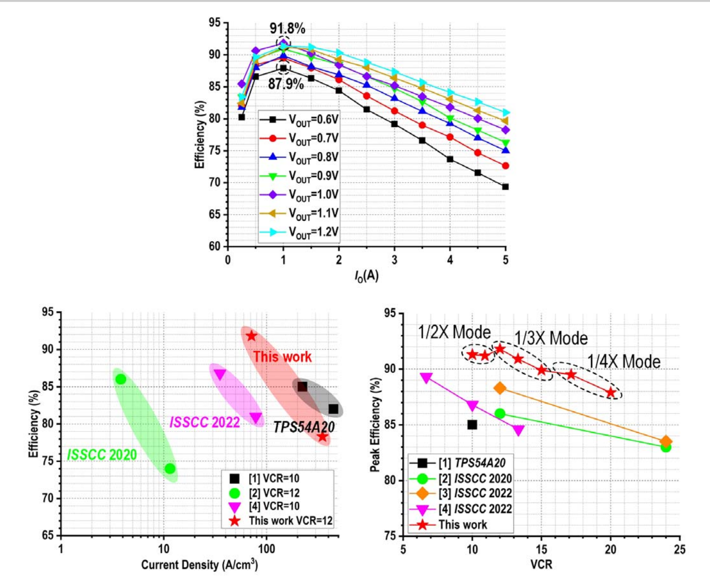


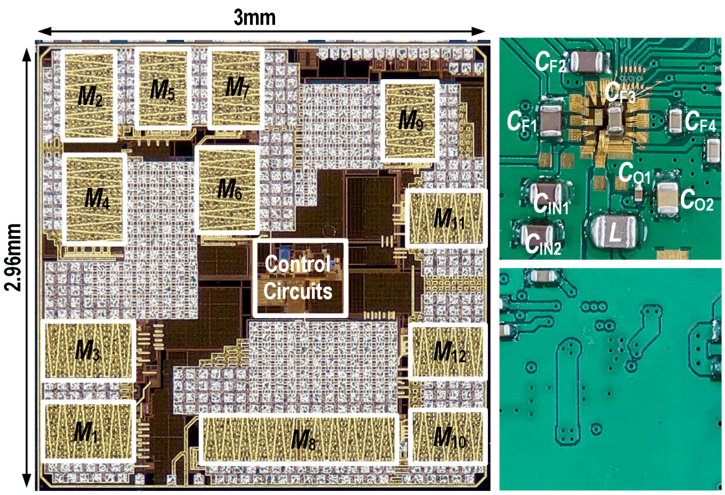
Figure 11.9.5: Measured efficiencies of the proposed ReSC-PL buck, and performance comparison with state-of-the-art works.

	[1] *TI* TPS54A20	[2] *ISSCC* 2020	[3] *ISSCC* 2022	[4] *ISSCC* 2022	**This Work**
Technology	N.A.	0.18μm BCD	0.18μm BCD	0.18μm HV SOI	**0.18μm BCD**
Topology	DSD	Tri-State DSD	DSD	CCC	**ReSC-PL**
Die area	N.A.	6.3mm²	9.6mm²	2.8mm²	**8.88mm²**
Input Voltage	12V	12/24V	12/24V	12V	**12V**
Output Voltage	1.2V	1V	1V	0.9-1.8V	**0.6-1.2V**
Maximum Output Current	10A	3A	4A	4A	**5A**
Switching Frequency	2MHz	1MHz	1MHz	2MHz	**0.8MHz**
Inductor	2×0.22μH	2×0.56μH	2×1.8μH	2×0.56μH	**1μH**
Inductor Size	3.2×2.5 ×1.2mm³	6.6×6.4 ×3.1*mm³	N.A.	3.5×3.2 ×2*mm³	**2.5×2 ×1.2mm³**
Flying Capacitor	2.2μF	2×1μF	4.7μF	2×2.2μF	**2×22μF 2×10μF**
Output Capacitor	94μF	10μF	10μF	10.6μF	**100μF**
Peak Efficiency @VCR	85@10	86*@12	88.3@12	86.8@10	**91.8@12**
Total Passive Volume†	0.0223cm³	0.262††cm³	N.A.	0.0511cm³	**0.0143cm³**
Maximum Current Density	448A/cm³	11.5A/cm³	N.A.	78.3A/cm³	**350A/cm³**

*Estimated from figure.

† Counts flying capacitors and inductors. †† Counts Inductors only.

Figure 11.9.6: Performance summary and comparison with state-of-the-art designs.



Component	Value	Size	Component	Value	Size
L	1µH	2.5×2×1.2mm³	C_{O1}	10nF	1×0.5×0.5mm³
C_{F1}, C_{F2}	22µF	2×1.25×1.25mm³	C_{O2}	100µF	2×1.25×1.25mm³
C_{F3}, C_{F4}	10µF	1.6×0.8×0.8mm³	C_{IN1}, C_{IN2}	10µF	2×1.25×1.25mm³

Figure 11.9.7: Chip micrograph and PCB of the proposed ReSC-PL buck converter.